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Thermal Demagnetization of some Carboniferous Lavas for Palaeomagnetic Purposes

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Summary

Thermal demagnetization of the natural magnetization of Carboniferous basalts from Scotland has successfully revealed the original direction of magnetization, although the specimens had badly scattered directions at room temperature.

1. Introduction

In two previous papers one of us (Wilson 1961, 1962a) has described experiments in which the changes in direction of the natural remanent magnetizations for various rock specimens were measured as a function of temperature. It was found that specimens which initially had widely scattered magnetizations yielded consistent directions when measured at high enough temperatures. These results were interpreted on the assumption that the heating removed various secondary components of magnetization, leaving only the original stable magnetization at the highest temperatures. The process was referred to as thermal cleaning.

In the earlier papers, only a few igneous rocks were examined, attention having been concentrated mainly on baked rocks. The present paper covers a systematic attempt to clean thermally a suite of British Carboniferous lavas, consisting of a series of about 30 basalt lavas with interbedded sediments of Calciferous Sandstone age, situated on the northern shore of the Firth of Forth, near Kinghorn, Fife. The strata are tilted around 30° in a north-easterly direction. A detailed description of the series has been given by Geikie (1900), who numbered the strata, sediments and lavas indiscriminately, from the bottom upwards; there being in all 69 numbered beds. His system of numbering is adopted here.

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Measurements of natural remanent magnetization at room temperature, for the lavas and also for a few baked sediments, had previously been carried out by C. W. F. Everitt. Consistent magnetic directions with small scatter occurred in about half the lavas and also in the sediments; whilst the remaining lavas showed very wide scatter. The mean axis of magnetization in the unscattered lavas was approximately 20° east of north and 27° up, relative to the original bedding planes. Along this axis both normal and reversed directions occurred; the normal and reversed lavas lying in successive groups. Most of these results, together with preliminary thermal cleaning measurements on a few lavas, were published by Everitt and Belshé (1960). In that paper several arguments were put forward mainly based on field evidence which suggested that the mean direction of 20° east, 27° up was the original field direction first recorded by these lavas.

The present paper includes a few of the earlier results, but almost all the present measurements were made by R. L. Wilson on new specimens cut from the original field samples. The specimens consisted of cylinders $\frac{7}{8}$-inch in diameter and about 1-inch long. Their natural magnetizations were measured as a function of temperature (Section 2) according to the procedure described by Wilson (1962b). The first specimens were generally measured every 50 °C, but at higher temperatures, 25 °C intervals were sometimes used. Later, when the technique was established, the specimens were preheated to 300 °C in a separate non-magnetic furnace in zero field, in order to speed up the measurements.

In an attempt to find possible differences between normal and reversed lavas, initial magnetization curves (up to 4 000 Oe) and Curie points (in 1 000 Oe) were determined. These results are described in Section 3.

Discussion and interpretation are in Section 4.

2. Effects of heating on the natural magnetization

The results are conveniently described in two parts. Sub-section 2.1 gives the collected general results for all specimens. Sub-section 2.2 describes in greater detail the results of thermal demagnetization of specimens from three separate lavas.

2.1. *Collected results*

Thermal demagnetization of the natural moments was carried out on thirty-three specimens from seventeen different Kinghorn lavas. All but three specimens were cut from independently oriented samples. The directions of magnetization almost always varied to some extent with temperature. The specimens fell into three classes according to their behaviour at high temperatures; the first class having single Curie points near 400–450 °C, the second having single Curie points near 580 °C, and the third class having both of these Curie points. The "high temperature" direction of magnetization may be defined as that direction existing less than about 100 °C below the temperature of final disappearance of the natural moment.

The room temperature directions for the seventeen lavas were widely scattered (Figure 1a), but the high temperature directions (Figure 1b) fell clearly into two antiparallel groups of directions, a normal group of nine lavas, and a reversed group of eight lavas. Directions of baked rocks are also indicated in Figure 1b. The mean axis of magnetization is 15·9° East of true North and 28·5° up (Table 1).

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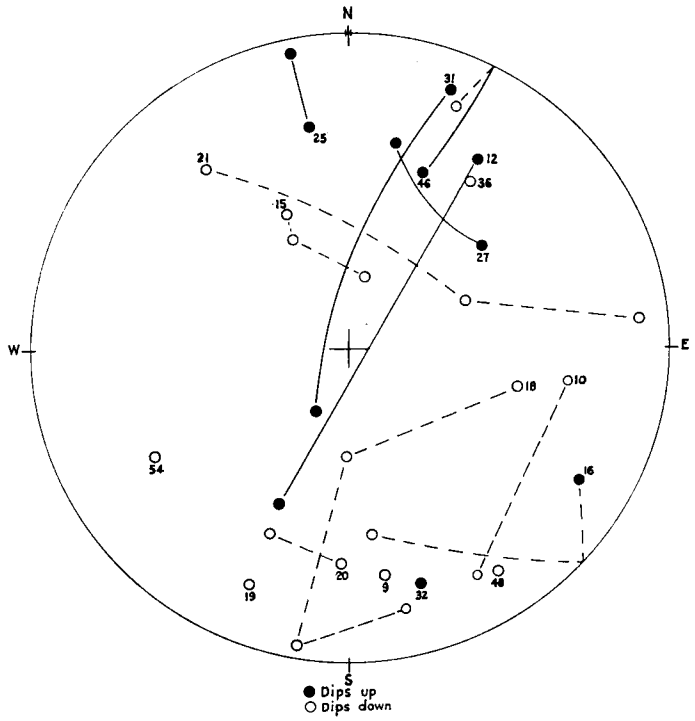


FIG. 1a.—The natural direction of magnetization of the thirty-three lava specimens from seventeen lavas, prior to heating.

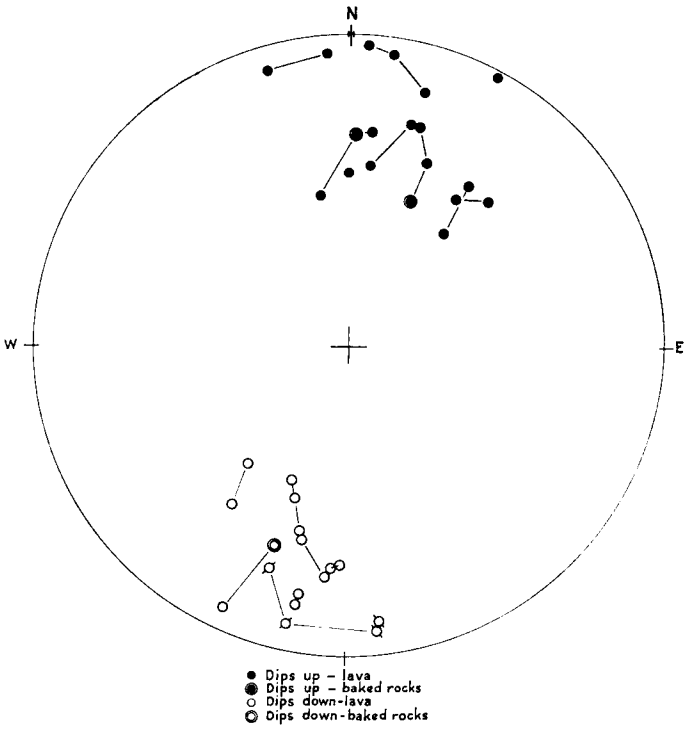


FIG. 1b.—The high temperature directions of magnetization of the same thirty-three specimens used for Fig. 1a. Baked specimens have been included, and are linked with their associated lavas.

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Table 1

Mean directions and deviations for the high temperature measurements of Table 5. Mean directions were found as usual by averaging the direction cosines of all the vectors in a group. The single reading standard deviation has been calculated considerating each specimen *to contribute one unit vector. For the deviation of the mean, each* lava *contributed a unit vector.*

Group	Mean declination	Mean inclination	Standard deviation for one reading	Standard deviation of the mean
Normal (9 flows)	15·4° east of true north	29·1° up	21°	7°
Reversed (8 flows)	16·4° +180° east of true north	27·9° down	15°	5·3°
Both considered as *normal* (17 flows)	15·9° east of true north	28·5° up	18°	4·4°

The standard deviation of directions is 18°, a range easily attributable to secular variation of the ancient field (Creer 1962). Details are recorded in Table 5 at the end of this paper.

2.2. *Individual results for three lavas*

Lava 15: This lava occurs about a quarter of the way up the series in a group of normally magnetized lavas (Table 5). Three separate field samples were available referred to here as 15A, B and C. The natural directions before heating were scattered over more than 90° (Table 2).

Table 2

Results for Flow 15

	Room temperature directions		High temperature directions	
	Declination (E of true N)	Inclination	Declination (E of true N)	Inclination
15A	41°	17° down	—	—
	281°	23° down	—	—
	5°	16° down	—	—
	280°	11° down	—	—
	346°	10° up	—	—
	333°	58° down	3°	3° up
15B	303°	50° up	—	—
	335°	51° down	16°	16° up
15C	22°	50° up	—	—
	11°	71° down	8°	6° up

The directions are given relative to the original bedding planes, and to true (geographic) north.

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Three cylinders, one from each sample, were measured during heating. The directions changed systematically as the temperature was raised until finally, above 370 °C, uniform residual magnetizations remained directed slightly east of north and upwards. These three final directions agreed well with each other (Table 2), and their mean value (9 °E and 8 ° up) fitted well with the results for the neighbouring lavas (Table 5 and Figure 1b) suggesting that the heating removed secondary magnetizations and also confirming the conclusion of Everitt

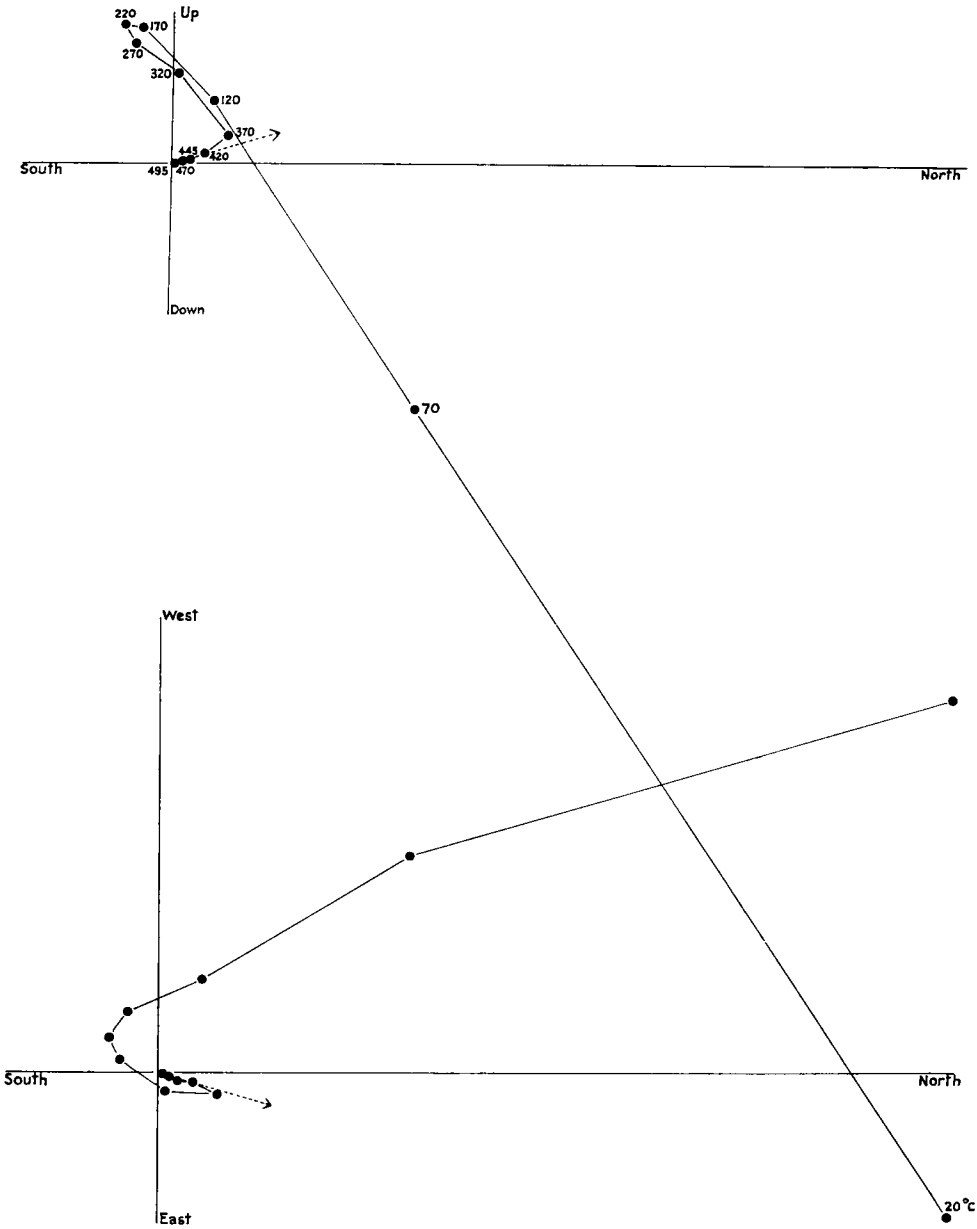


FIG. 2.—Vector heating diagram for specimen 15B, showing the change of magnitude and direction of natural magnetization with increasing temperature. The final high temperature direction is 16° East of True North and 16° above the ancient horizontal.

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and Belshé (1960) that the normal magnetizations in the neighbouring lavas were stable.

The vector heating curve for the specimen from 15B is given in Figure 2. First there was a magnetic moment which was erased at temperatures between 170 °C and 220 °C. Between 270 °C and 370 °C a second moment appeared, lying roughly opposite to the first. Finally, after a sharp break in direction just above 370 °C, a third moment appeared directed 16 °E and 16 ° up. This last weak moment is the only one which can be determined without ambiguity, because the *apparent* direction of the other two moments are influenced by the variation with temperature of the high temperature moment. Similar results, differing slightly in details, were obtained for the other two specimens.

Figure 3 shows the variation with temperature of *saturation magnetization* for a specimen of 15B, obtained in a field of 1 000 Oe. There is a well-defined Curie temperature between 370 °C and 420 °C corresponding to the second break

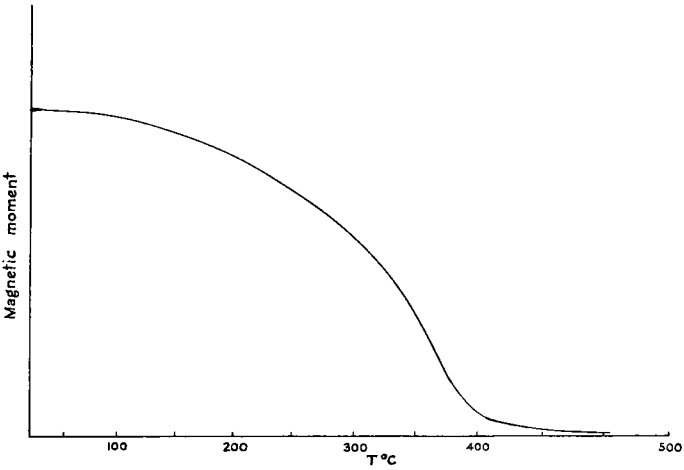


FIG. 3.—The moment of a specimen of 15B in 1 000 Oe in air, versus temperature.

in the vector heating curve (Figure 2) but on the other hand, there was no trace of any Curie point near 220 °C corresponding to the first break. Figure 3 also shows that the proportion of material with a Curie temperature above 400 °C was very small: its existence could hardly be established from this measurement alone, and was only clearly evident in the natural moment.

Lava 32: At room temperature the flows between 25 and 38 all showed normal (northward) magnetizations except for flow 32. Table 3 records the room temperature directions for specimens cut from the single available field sample of flow 32: although rather scattered, their general tendency is definitely southwards.

Figure 4 shows, in one plane, the vector heating curve for one specimen from flow 32. Once again as for the specimen from 15B there is a sudden change of direction, this time above 400 °C. The final high temperature direction is 359 °E, 43° up in good agreement with the neighbouring normal lavas. Thus the magnetization changed, on heating, from the southward polarity to the normal northward direction. However, this result does not represent self-reversal, since the two directions initially present in the rock are not even nearly antiparallel.

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Table 3

Results for Flow 32

Room temperature directions		High temperature directions	
Declination (E of true N)	Inclination	Declination (E of true N)	Inclination
185°	8° up	—	—
189°	8° down	—	—
221°	8° down	—	—
218°	16° down	—	—
187°	14° up	—	—
163°	23° up	359°	43° up

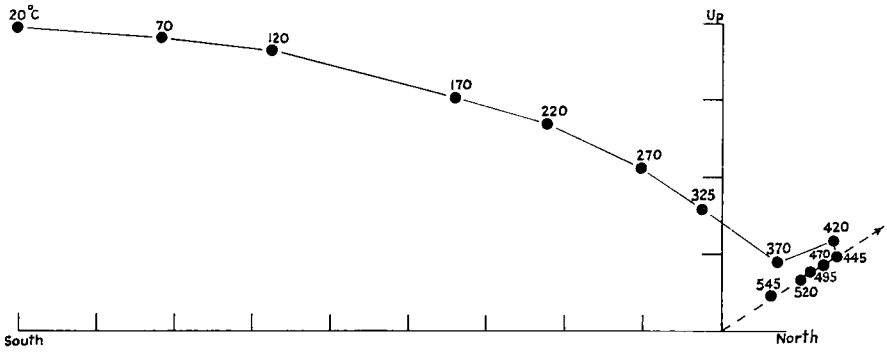


FIG. 4.—Vector heating diagram for specimen 32A, showing that its natural magnetization swings from south and up to north and up, this final direction being in agreement with the surrounding group of normal lavas (see Table 5 and Fig. 1b).

Lava 46: From flow 46 two field samples were available; 46A being from within two feet of the upper margin and 46B from near the centre of the flow (total thickness about twenty feet). The results from the two samples form an interesting contrast, since at room temperature each had some internal consistency, but the mean directions of the two blocks differed greatly (Table 4). However, at high temperatures the two directions coalesced and agreed with the room temperature direction for 46A (Table 4 and Figure 1b) as well as with an associated baked sediment. It seems, therefore, that 46A had retained its original magnetization while 46B had been altered magnetically after emplacement.

This interpretation fits with the magnetic measurements in Figures 5a, b and c, which show

5a heating in air in 1 000 Oe
5b heating of saturation remanence in zero field
5c alternating field demagnetization of saturation remanence

In Figure 5a, each specimen (46A and 46B) shows two Curie points at about 425 ° and 570 °C, respectively. But 46A has a much higher proportion of the high Curie point material and its saturation remanence has a greater resistance to

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both thermal demagnetization (Figure 5b) and alternating field demagnetization (Figure 5c). It is customary in palaeomagnetic research to regard these three features as characteristic of a magnetically stable material. Since the vector heating measurements also indicated that 46A was probably stable, the customary interpretation appears to be justified in the present instance at least.

Table 4

Results for Flow 46

	Room temperature measurements		High temperature measurements	
	Declination (E of true N)	Inclination	Declination (E of true N)	Inclination
46A	25°	37° up	—	—
	27°	41° up	—	—
	22°	37° up	—	—
	18°	38° up	—	—
	21°	41° up	—	—
	21°	42° up	—	—
	22°	39° tup	22°	36° up
46B	80°	22° down	—	—
	67°	48° down	—	—
	23°	17° down	17°	27° up
Sandstone baked by Flow 46	19°	49° up	—	—
	25°	48° up	—	—

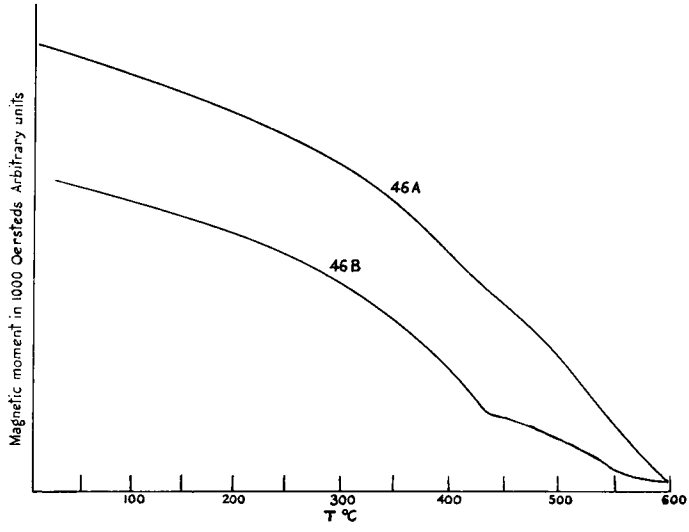


FIG. 5a.—Magnetic moment in 1000 Oe in air, versus temperature, for specimens 46A and 46B.

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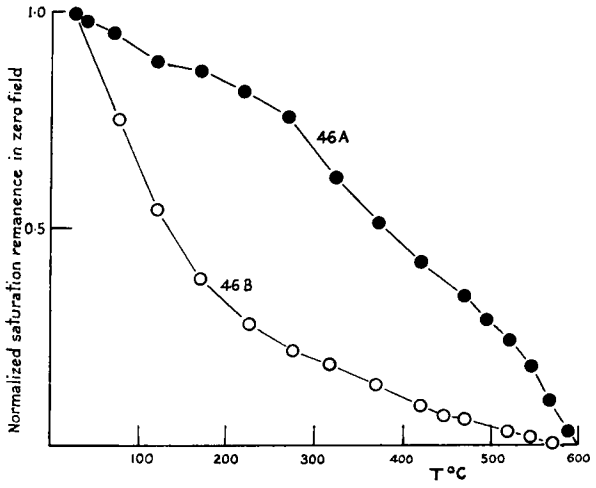


FIG. 5b.—Saturation remanence in zero field in air, versus temperature, for specimens 46A and 46B.

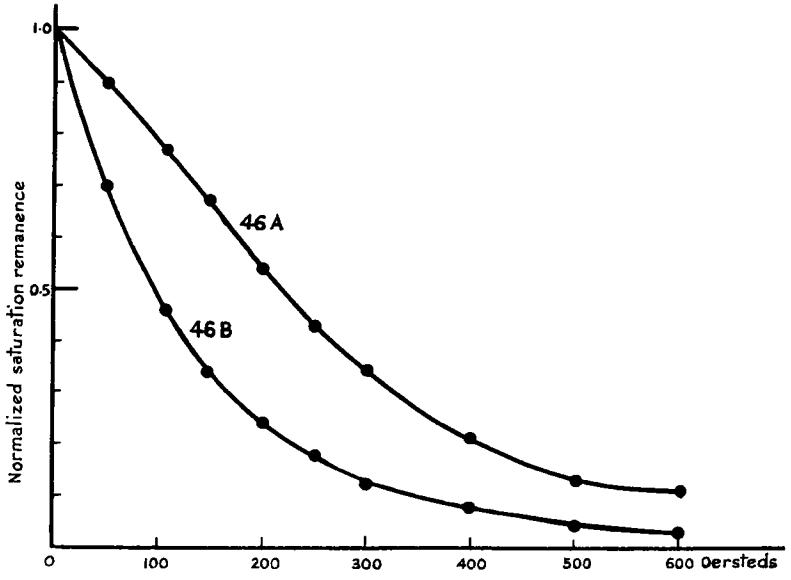


FIG. 5c.—Alternating field demagnetization of specimens 46A and 46B.

3. General examination of magnetic properties

Two types of measurement were performed to try to discern any possible physical differences between normal and reversed lavas. Initial magnetization curves were plotted for fields up to 4 000 Oe, and thermal demagnetizations were done in a 1 000 Oe field. Figure 6 shows that the distribution of saturation magnetizations is the same for both groups. So were the shapes of the initial magnetization curves themselves. The Curie points of eleven specimens are listed in Table 5, and neither these nor the shapes of the thermal demagnetization curves revealed any distinction between normal and reversed lavas.

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Table 5
Summary of Results

Bed	Description	Thickness	No. of field samples	Low Temperature Directions: No. of specimens	Mean declination	Mean inclination	Scatter	High Temperature Direction: No. of specimens	Mean declination	Mean inclination	Scatter	Normal or reversed	Remarks	Curie points
1–8	Sediments		[not sampled]	—	—	—	—	—	—	—	—	—	—	—
9	Basalt	1	1	4	80°	40° down	80°	1	191°	16° down	—	R		
10	Basalt		1	4	180°	70° down	60°	2	189°	32° down	~4°	R		460°
11	Black shale	6 in.	[not sampled]	—	—	—	—	—	—	—	—	—	—	—
12	Basalt		2	2	37°	30° up	20°	2	11°	34° up	~16°	N	Low temperature measurements on blocks not small specimens	
13	Basalt		[not sampled]	—	—	—	—	—	—	—	—	—	—	—
14	Shale and sandstones	12 ft	[not sampled]	—	—	—	—	—	—	—	—	—	—	—
15	Basalt		3	3	343°	28° up	30°	3	9°	8° up	~5°	N	Low temperature measurements on blocks only (compare Table 1). Layer corresponds to a break in the succession	400°
15A	Sediments	50 ft	[not sampled]	—	—	—	—	—	—	—	—	—	—	—
16	Basalt		2	6	201°	25° down	35°	2	218°	43° down	~6°	R	Low temperature results from one field sample only	450°
17	Shale and limestone		[not sampled]	—	—	—	—	—	—	—	—	—	—	—
18	Basalt		2	8	169°	8° down	6°	4	185°	13° down	~10°	R	—	—
19	Basalt		3	9	176°	21° down	30°	1	191°	20° down	—	R	—	410°, 500°
20	Basalt	3–6 ft	2	8	182°	35° down	30°	2	183°	29·5° down	1°	R	—	—

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21	Basalt 20 ft	3	12	[no consistent direction]			3	198°	45° down	~6°	R	—	—
22	Green tuff	[not sampled]	—	—	—	—	—	—	—	—	—	—	—
23	Black shale	[not sampled]	—	—	—	—	—	—	—	—	—	—	—
24	Green sandy tuff	1	—	—	—	—	—	—	—	—	—	Not measured —	—
25	Basalt	3	8	356°	17° up	~8°	2	349°	7° up	~6°	N	Results from two field samples only —	
26	Black shale and tuff	[not sampled]	—	—	—	—	—	—	—	—	—	—	—
27	Basalt	3	9	18°	32° up	12°	2	38°	44° up	~8°	N	—	380°
28	Black shale	[not sampled]	—	—	—	—	—	—	—	—	—	—	—
29	Sandy and ashy limestone	[not sampled]	—	—	—	—	—	—	—	—	—	—	—
30	Baked green shale	1	12	1°	32° up	7°	—	—	—	—	N	—	—
31	Basalt baking	2	8	356°	5° up	35°	2	357°	40°up	~9°	N		430°, 580°
32	Basalt	1	4	198°	6° down	20°	1	359°	43° up	—	N		
33	Basalt	3	5	20°	39° up	9°	—	—	—	—	—	Results for one field sample only, second sample widely scattered	
34	Basalt	2	8	—	—	—	—	—	—	—	—	Very wide scatter before heating	480°
35	Tuffaceous shaly sandstone	1	12	5°	39° down	17°	—	—	—	—	?	Probably unstable	
36	Basalt	3	20	—	—	—	1	28°	2° up	—	N	Very wide scatter before heating	

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Table 5—*continued*

Bed	Description	Thickness	No. of field samples	Low Temperature Directions: No. of specimens	Mean declination	Mean inclination	Scatter	High Temperature Directions: No. of specimens	Mean declination	Mean inclination	Scatter	Normal or reversed	Remarks	Curie points
37	Baked tuff		1	—	—	—	—	—	—	—	—	—	Too friable to measure	
38	Basalt		3	20	—	—	—	—	—	—	—	—	Very wide scatter	
39	Basalt		1	4	270°	50°	60°	—	—	—	—	?		
40	Red tuff		1	9	—	—	—	—	—	—	—	—	Directions changed when specimens left in earth's field for a few hours	
41	Basalt baking	40	1	6	36°	39° up	4°	—	—	—	—	N		
42	Sediments		[not sampled]	—	—	—	—	—	—	—	—	—		
43	Basalt		3	6	21°	42° up	10°	—	—	—	—	N	There may be two lavas in this	
43a	Baked tuff		1	6	22°	71° up	7°	—	—	—	—	N	group. Measurements are on one field sample only in lower half of group	
44	Basalt		1	12	—	—	—	—	—	—	—	—	Very wide scatter	
45	Sandstones	200 ft	2	2	22°	48° up	2°	—	—	—	—	N N	Baked specimens taken from top of 200 foot section	
46	Basalt	20 ft	2	6	22°	39° up	3°	2	20°	32° up	~5°	—	Low temperature measurements taken from one field sample only	420, 580
47	Shale or marl		[not sampled]	—	—	—	—	—	—	—	—	—	—	
48	Basalt		3	12	—	—	—	2	40°	39° up	4°	N	Low temperature measurements showed wide scatter	430, 570
49	Sandstone		1	—	—	—	—	—	—	—	—	—	Too weak to measure	
50	Basalt		1	6	191°	40° down	10°	—	—	—	—	R		
51	Sediments		[not sampled]	—	—	—	—	—	—	—	—	—		

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52a	Basalt (lowest flow)	1	6	205°	56° down	6°	—	—	—	—	R	These three samples all from the 300 yards series of lavas grouped together by Geikie. 52c measured on complete field sample. Individual specimens showed wide scatter
52b	Baked sandstone	1	—	—	—	—	—	—	—	—	—	
52c	Basalt (top flow)	1	1	260°	62° down	—	—	—	—	—	—	
53	Baked sandstone	1	12	200°	32° down	5°	—	—	—	—	R	At contact with basalt 54
54	Basalt	1	12	243°	23° down	15°	1	205	8° down	—	R	
55	Basalt	2	16	192°	27° down	30°	—	—	—	—	R	
56–61	Sediments	[not sampled]	—	—	—	—	—	—	—	—	—	—
62	Calcareous shale	1	—	—	—	—	—	—	—	—	—	Too weak to measure
63	Basalt	1	4	—	—	—	—	—	—	—	—	Very wide scatter
64	Red tuff	2	15	34°	32° down	25°	—	—	—	—	N?	
65	Basalt	2	18	15°	14° down	9°	—	—	—	—	N?	Possibly a sill
66–67	Sediments	[not sampled]	—	—	—	—	—	—	—	—	—	
68	Baked sandstone	1	—	—	—	—	—	—	—	—	—	Wide scatter
69	Basaltic sill	2	—	15°	35°	—	—	—	—	—	—	Wide scatter

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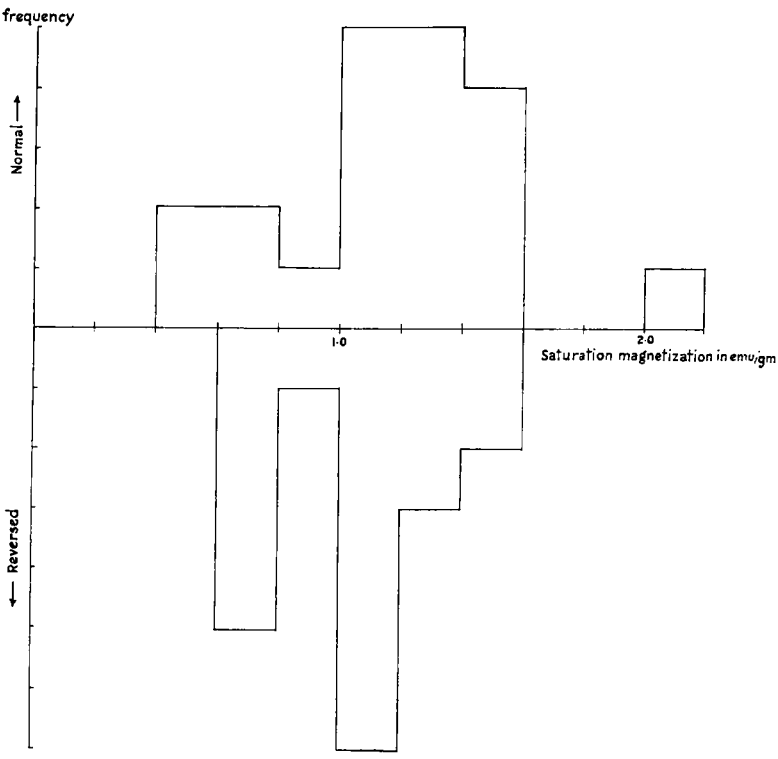


FIG. 6.—Histograms of magnetic moments in 3500 Oe, for groups of both normal and reversed rocks.

4. Discussion

Comparison of the results in Figures 1a and 1b suggests strongly that in this instance the original palaeomagnetic directions in specimens with wide scatter were recovered by measurements at high temperatures. The mean direction of 15·9 °E, and 28·5° up agrees well with the Carboniferous direction of 20 °E,27° up arrived at by Everitt and Belshé (1960). Reference should be made to that paper for a discussion of field tests for stability and also for an analysis of certain other apparently conflicting results in rocks of Carboniferous age. The axis of magnetization also agrees well with those from the British Carboniferous exposures at Clee Hill (200° East and North, 15° down) and at Tideswelldale (218°, 36° down) where baked contact materials were available as corroborating evidence (Everitt, 1959 and 1960). Other British Carboniferous rocks agree less well (Clegg *et al.* 1954, Blundell 1957, Everitt 1960, Belshé 1957, Everitt and Belshé 1960).

Assuming that the Earth's mean field was then an axial dipole, the results as a whole indicate that in Carboniferous times Britain was about 15° south of the equator and was rotated 16° counterclockwise from its present attitude.

While this interpretation appears convincing, it raises interesting questions on how to explain the various components of magnetization revealed by the heating experiments.

The low temperature components, which disappear around 200 °C, are probably due to viscous remagnetization of some of the material (Thellier 1951) since when they occur (e.g. Figure 1) they appear to lie roughly along the Earth's present field direction.

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The higher temperature components are more complicated. Three types of specimens existed distinguished by

(a) a single Curie point between 400–450 °C,

(b) a single Curie point near 580 °C,

(c) two Curie points, at 400–450 °C and about 580 °C respectively.

For all lavas with single Curie points the magnetization fell into the pattern of Figure 1b in each case at least within about 100 °C of the Curie point. On the other hand in lavas with double Curie points the magnetization of the higher Curie point material fitted the pattern, but the magnetization associated with the lower Curie point was anomalous, as can be seen for example in Figures 2 and 4. Thus anomalous directions were associated with a Curie point around 400–450 °C, but occurred only in specimens which also contain some higher Curie point material.

This curious result is difficult to explain. We have considered the following hypotheses:

(*a*) that the minerals now present existed at the time of formation, but that when the two different Curie point materials coexist, the one with the lower Curie point can somehow be re-magnetized fairly easily. Experiments by Mussett (1961) demonstrate that in a basalt from India with a double Curie point, the lower Curie point material, but not the higher, became remagnetized under pressure applied in the laboratory. There is therefore some evidence for instability of a lower Curie point material when associated with a higher;

(*b*) that in the lavas with a single low Curie point of 450 °C, the magnetic mineral is original; but that in the lavas with double Curie points, the 450 °C material was produced by an incomplete mineralogical transformation long after formation, and therefore has a later direction of magnetization;

(*c*) that in the double Curie point lavas the two minerals interact magnetically so that the lower Curie point material becomes magnetized at an angle to the upper one and therefore at an angle to the ambient field.

The second and third explanations both seem highly improbable, so that we consider the first explanation to be the most likely.

The reversed natural moments are probably due to reversal of the Earth's magnetic field, as suggested by Everitt and Belshé (1959). The main grounds for this conclusion are that:

(*a*) concordant baked rocks exist in conjunction with both normal and reversed lavas;

(*b*) normal and reversed lavas occur in well defined groups;

(*c*) laboratory tests on saturation magnetization and Curie points show no systematic magnetic differences between normal and reversed specimens.

Test (*a*) is quite conclusive, but tests (*b*) and (*c*), though widely accepted as valid, must be treated with caution, since Ade-Hall and Wilson (1963) have found clear *petrological* differences between certain normal and reversed lavas while no magnetic differences were found.

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